COMP 2121 DISCRETE MATHEMATICS — LAB 11

<u>Merge Algorithm</u>

Step 1: Set L equal to the empty list.

Step 2: Compare the first elements in L_1, L_2. Remove the smaller of the two from the list it is in and place it at the end of L.

Step 3: If either of L_1, L_2 is empty then the other list is concatenated to the end of L else go to step 2.

Theorem. Let L_1 and L_2 be two sorted lists of ascending numbers, where L_1 contains n_1 elements and L_2 contains n_2 elements. Then L_1 and L_2 can be merged into one ascending list L using at most $n_1 + n_2 - 1$ comparisons.

1. Give an example of two lists L_1, L_2, each of which is in ascending order and contains 6 elements, and where a) 11 comparisons, b) nine comparisons, c) 6 comparisons, are needed to merge L_1, L_2 by the Merge Algorithm.

a) 11 comparisons

$L_1 : \cancel{1}\ \cancel{3}\ \cancel{5}\ \cancel{7}\ \cancel{9}\ \cancel{11}$

$L_2 : \cancel{2}\ \cancel{4}\ \cancel{6}\ \cancel{8}\ \cancel{10}\ 12$ copied

$L : \underbrace{1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11}_{\text{Compared}}\ 12$

b) 9 comparisons

$L_1 : \cancel{1}\ \cancel{3}\ \cancel{5}\ \cancel{7}\ \cancel{8}\ \cancel{9}$

$L_2 : \cancel{2}\ \cancel{4}\ \cancel{6}\ 10\ 11\ 12$ Copied

$L : \underbrace{1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9}_{\text{compared}}\ 10\ 11\ 12$

c) 6 comparisons

$L_1 : \cancel{1}\ \cancel{2}\ \cancel{3}\ \cancel{4}\ \cancel{5}\ \cancel{6}$

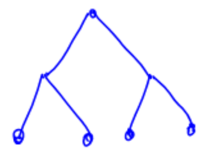
$L_2 : 7\ 8\ 9\ 10\ 11\ 12$ copied

$L : \underbrace{1\ 2\ 3\ 4\ 5\ 6}_{\text{compared}}\ 7\ 8\ 9\ 10\ 11\ 12$

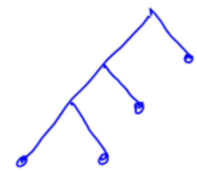
2. Let L_i, for $1 \leq i \leq 4$, be four lists of numbers, each sorted in ascending order. The number of entries in these lists are 75, 40, 110, and 50, respectively.

a) In how many different ways lists $L_1,\ L_2,\ L_3$ and L_4 can be merged into a single list?

Two possible trees



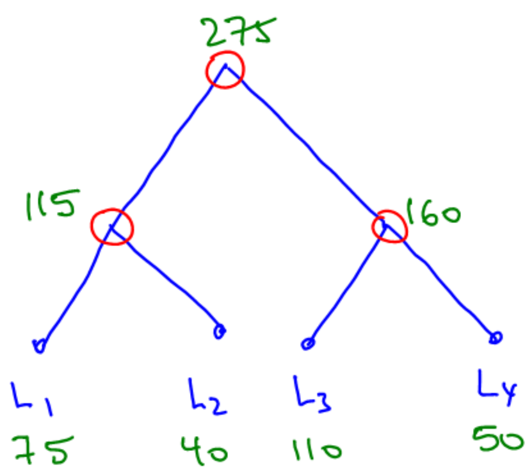
or



$$\frac{\binom{4}{2}}{2} + \binom{4}{2}\cdot(2)(1) = 3+12 = 15$$

b) What is the most number of total comparisons required to merge these four lists by merging L_1 and L_2, merging L_3 and L_4, and then merging the two results? What is the least number of total comparisons required by the given pattern?

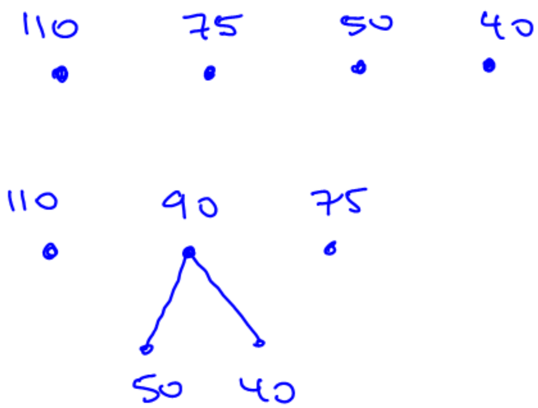
One of the 15 trees

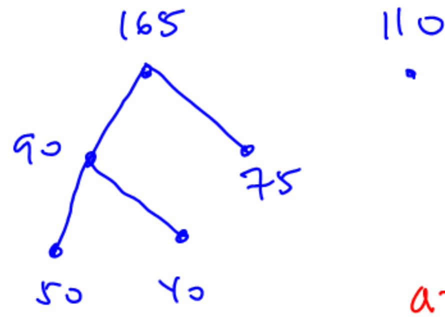


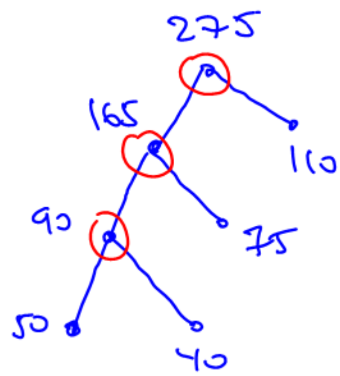
at most: $(115-1)+(160-1)+(275-1) = 547$

at least: $40+50+115 = 205$

c) Utilize Huffman Algorithm and find the optimal merge pattern that minimizes the total number of needed comparisons. What is the most number of comparisons required by the merge pattern that you presented? What is the least number of comparisons required by the merge pattern that you presented?







at most: $(275-1) + (165-1) + (90-1) = 527$

at least: $40 + 75 + 110 = 225$

3. Let A and B be two lists sorted in ascending order, where A has m elements, B has n elements. Under what condition, <u>exactly</u> $m+n-1$ comparisons are needed to merge A and B by the Merge Algorithm?.

$A : a_1, a_2 \dots a_{m-1}\, a_m$

$B : b_1, b_2 \dots b_{n-1}\, b_n$

To have exactly $m+n-1$ comparisons, the largest two elements need to be in two diffent lists.

expected # of comparisons (average case):

$(0.5)(m+n-1) + (0.25)(m+n-2) + (0.125)(m+n-3) + \dots = m+n-2$

algebra not covered in this course

A Few Review Problems:

1. Derive an explicit formula (dependent on n) for the value returned by the following function. Then prove your formula using mathematical induction.

```
MyFunction (positive integer n)

    i = 1
    j = 2
    while i <= n do
        j = 3j - 2
        i = i + 1
    end while

return j
```

n	Mystery Funct. returns	non-recursive formula $j_n = 3^n + 1$	3^n
1	$3(2)-2=4$	$3^1+1=4$	3
2	$3(4)-2=10$	$3^2+1=10$	9
3	$3(10)-2=28$	$3^3+1=28$	27
4	$3(28)-2=82$	$3^4+1=82$	81

The function returns $j_n = 3^n + 1$; $n \geq 1$

① Base case: verify that the statement is true for $n=1$

code: For $n=1$ the while loop runs once. When $n=1$ then $i=1, j=2$. The statement $i \leq n$ is $1 \leq 1$ and the while loop runs $j = 3(2) - 2 = 4$. The while loop exists, function returns $j = 4$

The same; verified

formula: $j_1 = 3^1 + 1 = 4$

② Assume that the folloing is true for $n = k$

The function returns $j_k = 3^k + 1$ ← use this!

③ Target, for $n = k+1$

The function returns $j_{k+1} = 3^{k+1} + 1$

④ Proof

When $n = k+1$ is passed to the function, in the last iteration of the while loop

$$j_{k+1} = 3 \cdot j_k - 2$$

step 2

$$= 3 \cdot (3^k + 1) - 2$$

$$= 3 \cdot 3^k + 3 - 2$$

$a^x \cdot a^y = a^{x+y}$

$$= 3^{1+k} + 1 = 3^{k+1} + 1$$

(matches the target)

2. Find the value of the variable counter (in terms of n) after the following programming segment is executed:

```
counter = 300;
for (i = 1; i <= n; i++){
      for (j = 10; j <= i+1; j++){
            counter = counter + 8j;

      }
      counter = counter + 7;
}
```

does not run for $i = 1, 2, \ldots, 8$

Your formula must be in the simplest form.***In order to receive marks your approach must rely on sigma notation, summation properties and usage of given formula(s). Answers based on combinatorics will not be accepted.

You may use the following formulas: $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$, $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$

$$\text{counter} = 300 + \sum_{i=9}^{n}\sum_{j=10}^{i+1} 8j + \sum_{i=1}^{n} 7 = 300 + \sum_{i=9}^{n}\left[8\sum_{j=10}^{i+1} j\right] + (n-1+1)\cdot 7$$

!!

need to start at 1 to use formula

$$= 300 + \sum_{i=9}^{n}\left[8\left(\sum_{j=1}^{i+1} j - \sum_{j=1}^{9} j\right)\right] + 7n = 300 + \sum_{i=9}^{n}\left[\cancel{8}^{4}\cdot\left(\frac{(i+1)(i+2)}{\cancel{2}} - \frac{9(10)}{\cancel{2}}\right)\right] + 7n$$

$$= 300 + \sum_{i=9}^{n}\left[4\left(i^2+2i+i+2-90\right)\right] + 7n = 300 + \sum_{i=9}^{n}\left(4i^2+12i-352\right) + 7n$$

$$= 300 + 4\sum_{i=9}^{n} i^2 + 12\sum_{i=9}^{n} i - \sum_{i=9}^{n} 352 + 7n$$

need to start at 1

count! no need for i=9

$$= 300 + 4\left[\sum_{i=1}^{n} i^2 - \sum_{i=1}^{8} i^2\right] + 12\left[\sum_{i=1}^{n} i - \sum_{i=1}^{8} i\right] - (n-9+1)(352) + 7n$$

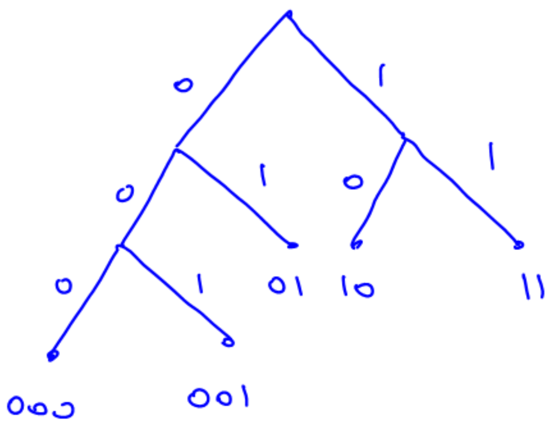
$$= 300 + \cancel{4}^{2}\left[\frac{n(n+1)(2n+1)}{\cancel{6}\,3} - \frac{8(9)(17)}{\cancel{6}\,3}\right] + \cancel{12}^{6}\left[\frac{n(n+1)}{\cancel{2}} - \frac{8(9)}{\cancel{2}}\right] - 352n + 8(352) + 7n$$

$$= 300 + \frac{2}{3}n(2n^2+3n+1) - 816 + 6\left[n^2+n-72\right] - 352n + 2816 + 7n$$

$$= 300 + \frac{4}{3}n^3 + 2n^2 + \frac{2}{3}n - 816 + 6n^2 + 6n - 432 - 352n + 2816 + 7n$$

$$= \frac{4}{3}n^3 + 8n^2 - \frac{1015}{3}n + 1868$$

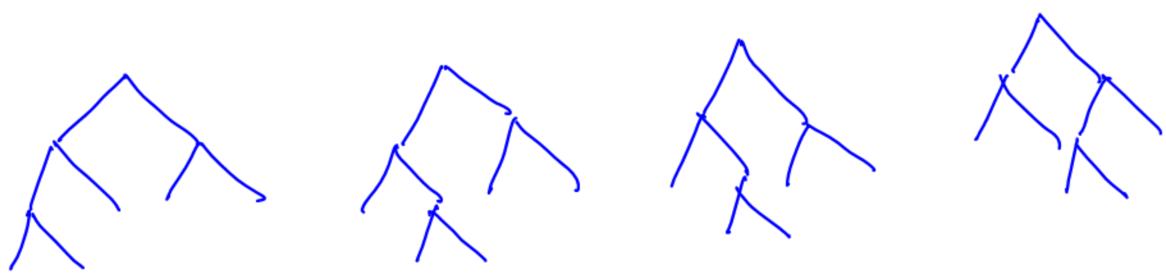
3. A text file is composed only of the letters A, B, C, D and E. In how many different ways can we encode the file using codewords with the prefix property that have exactly 3 codewords of length 2 and 2 codewords of length 3?



000
001
01
10
11

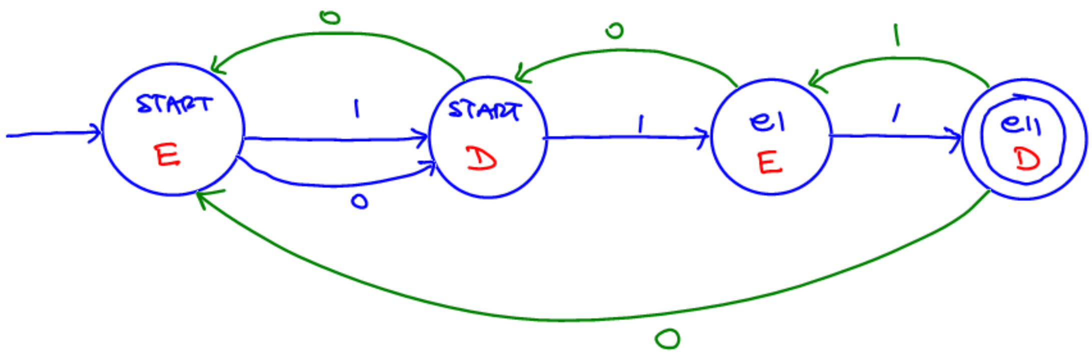
can assign A, B, C, D, E in 5! ways

4 trees

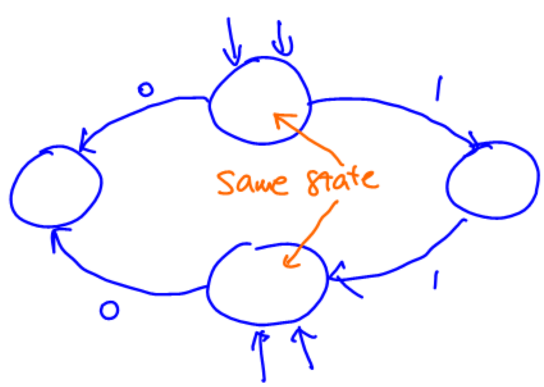


$\therefore 4(5!)$

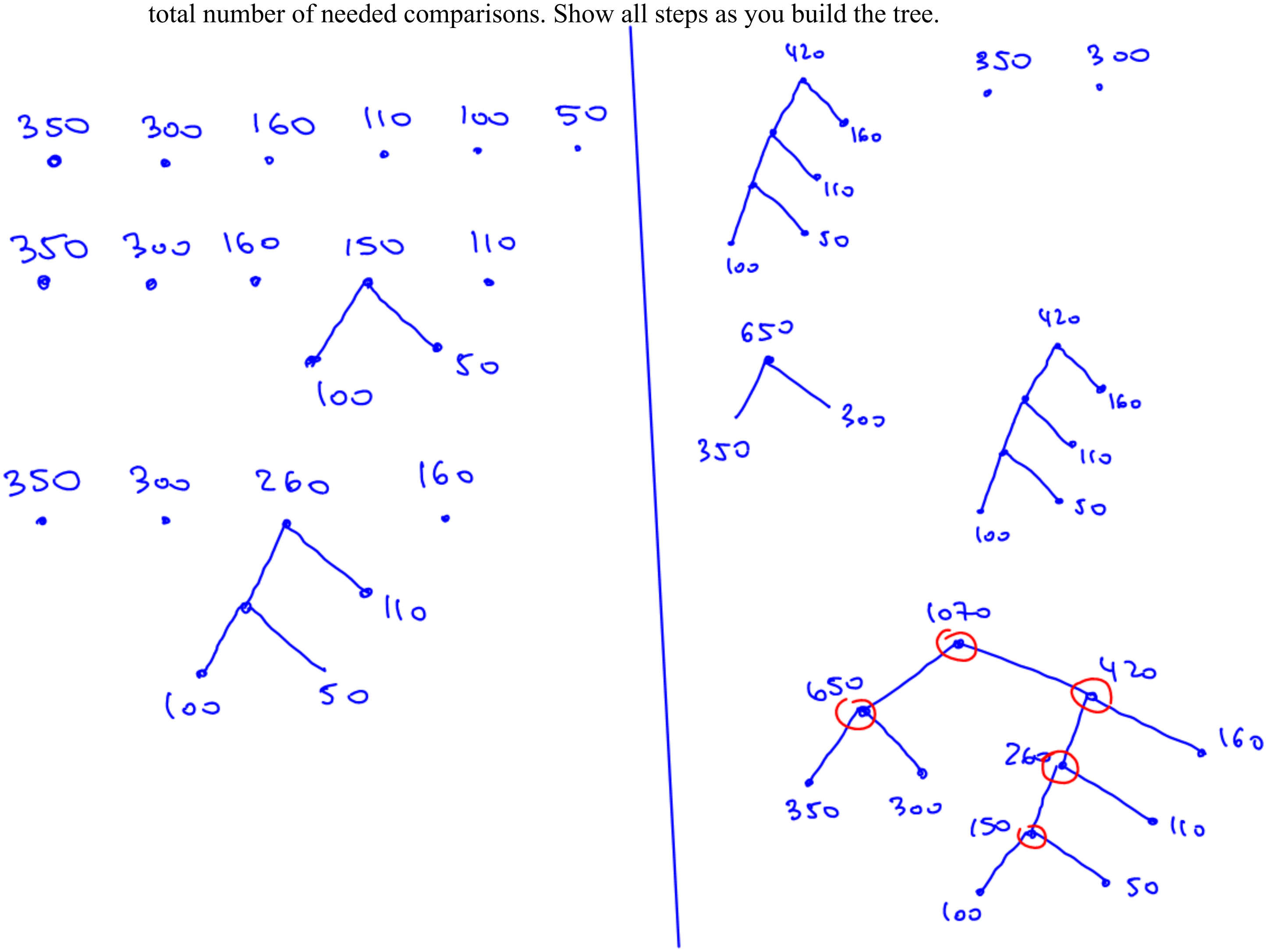
4. Design a finite state machine that recognizes (accepts) all binary strings that end in 11 and have odd length.



watch for extra
states



5. (a) We wish to merge six lists with 50, 100, 110, 160, 300 and 350 numbers (each sorted in ascending order) into a single list sorted in ascending order. By merging only two lists at the time and utilizing Huffman Algorithm, find the optimal merge pattern that minimizes the total number of needed comparisons. Show all steps as you build the tree.



(b) What is the most number of comparisons required by the merge pattern that you presented in part(a)? What is the least number of comparisons required by the merge pattern that you presented in part(a)?

at most : $(650-1)+(1070-1)+(420-1)+(260-1)+(150-1)$

$= 2545$

at least : $300+420+160+110+50$

$= 1040$